

Alden Bernstein

a2bernstein@ucsd.edu — <https://aldenbernsteinm.github.io/Profile/> — <https://www.linkedin.com/in/alden-bernstein-31b1a22b0/>

EDUCATION

University of California San Diego

Bachelor of Science (B.S) in Computer Science

San Diego, CA

Expected May 2026

Relevant Coursework:

- Data Structures & Algorithms, Object-Oriented Design, Distributed Systems
- Database Management, Machine Learning, Data Science

WORK EXPERIENCE

Software Engineer Intern

University of California San Diego

Dec 2024 – Present

San Diego, CA

- Researching and applying PyTorch/TensorFlow models to accelerate research document analysis for UCSD departments
- Built a feedback system using PostgreSQL reducing model hallucinations by 30% via A/B testing
- Implemented Grafana analytics pipeline for 25% faster model optimization

Software Engineer Intern

University of California San Diego

Oct 2024 – Jan 2025

San Diego, CA

- Built a data validation web tool with React.js for the Scripps Institution
- Streamlined the validation process by automating an hours-long manual workflow, reducing processing time by 95%
- Preprocessed datasets using javascript to improve anomaly detection accuracy by 25%

Software Engineer Intern

Crunch Digital

Aug 2024 – Oct 2024

San Diego, CA

- Implemented RESTful API integrations for client-server music applications using Python, FastAPI, and AWS
- Identified and resolved misconfigured endpoint parameters and inefficient data serialization, reducing errors by 35%
- Collaborated with a 5-person team to test and deploy a reporting API, automating licensing reports for 10+ music stakeholders

PROJECTS

Protocol Pilot AI — Python, Pytorch, Tensorflow, LangChain, ChromaDB

Developed a predictive model for compliance verification between guidelines and documents under review using LSTM-based Recurrent Neural Networks Leveraged Python and TensorFlow to build the model, utilizing NLP techniques to process and analyze textual data for compliance verification.

RAG-Powered Semantic Indexing — Python, Tensorflow, ChromaDB, NumPy, Pandas

Developed a Retrieval-Augmented Generation (RAG) system using an optimized LLAMA model that enables users to analyze their embedded data at no cost on their local machine. Additionally, implemented a multi-session chat archive, which taught me how to effectively continue LLM conversations without relying on external libraries.

UCSD Minor Code Search — Python, NumPy, Pandas, PHP

Implemented an online demo with PHP for real-time minor catalog searches, consolidating all necessary course information into one accessible platform. This solution addressed a common pain point for UCSD students, streamlining the process of course planning and ensuring they could easily find the relevant information needed to create their academic schedule

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript/TypeScript, HTML, CSS, SQL, Go

Libraries & Frameworks: React.js, Node.js, JUnit, Pytest, PyTorch, Tensorflow, Flask

Tools & Technologies: Git, Docker, AWS (Lambda, S3, EC2), Kubernetes

Databases: AWS DynamoDB, MySQL, PostgreSQL, MongoDB